www.tntextbooks.in

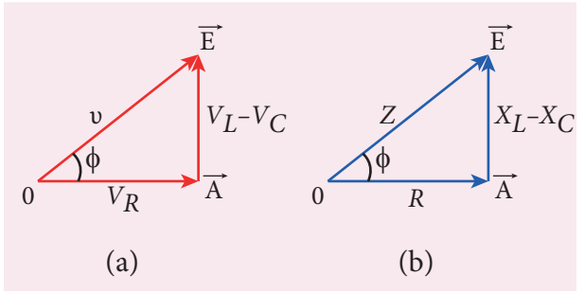


Figure 4.53 Voltage and impedance triangle when $X_L > X_C$

From phasor diagram, the phase angle between v and i is found out from the following relation

$$\tan\phi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R} \qquad (4.55)$$

Special cases

(i) If $X_L > X_C$, $(X_L - X_C)$ is positive and phase angle ϕ is also positive. It means that the applied voltage leads the current by ϕ (or current lags behind voltage by ϕ). The circuit is inductive.

$$\therefore \upsilon = V_m \sin\omega t;\ i = I_m \sin(\omega t - \phi)$$

(ii) If $X_L < X_C$, $(X_L - X_C)$ is negative and ϕ is also negative. Therefore current leads voltage by ϕ and the circuit is capacitive.

$$\therefore \upsilon = V_m \sin\omega t;\ i = I_m \sin(\omega t + \phi)$$

(iii) If $X_L = X_C$, ϕ is zero. Therefore current and voltage are in the same phase and the circuit is resistive.

$$\therefore \upsilon = V_m \sin\omega t;\quad i = I_m \sin\omega t$$

4.7.7 Resonance in series RLC Circuit

When the frequency of the applied alternating source (ω_r) is equal to the natural frequency $\left[1/\sqrt{LC}\right]$ of the *RLC* circuit, the current in the circuit reaches its maximum value. Then the circuit is said to be in electrical resonance. The frequency at which resonance takes place is called resonant frequency.

Resonant angular frequency, $\omega_r = \dfrac{1}{\sqrt{LC}}$

or $$f_r = \frac{1}{2\pi\sqrt{LC}} \qquad (4.56)$$

At series resonance,

$$\omega_r = \frac{1}{\sqrt{LC}} \quad \text{or} \quad \omega_r^2 = \frac{1}{LC}$$

$$\omega_r L = \frac{1}{\omega_r C} \quad \text{or}$$

$$X_L = X_C \qquad (4.57)$$

Table 4.1 Summary of results of AC circuits

Type of Impedance	Value of Impedance	Phase angle of current with voltage	Power factor
Resistance	R	0°	1
Inductance	$X_L = \omega L$	90° lag	0
Capacitance	$X_C = 1/\omega C$	90° lead	0
R- L - C	$\sqrt{R^2 + \left(\omega L - 1/\omega C\right)^2}$	Between 0° and 90° lag or lead	Between 0 and 1

UNIT-4(XII-Physics_Vol-1).indd 261 04-03-2019 10:41:12